

Data Structures Study Guide

Coding Interview Prep — Basic, Collection & Advanced Structures

1. Basic Data Structures

Fundamental data types that store single values or basic sequential collections.

- **Integers & Booleans:** Primitive types representing whole numbers and logical truth states (true/false).
- **Strings:** Sequences of characters used for textual representation and manipulation.
- **Arrays:** Sequential blocks of contiguous memory providing $O(1)$ constant-time element lookup by index.
- **Objects:** Key-value collections mapping property names to dynamic values.

2. Collection Data Structures

Structures designed to hold, organize, and manage groups of elements with specific access patterns.

Structure	Access Pattern / Behavior	Primary Use Case
Lists & Sets	Lists store ordered sequences; Sets enforce unique elements.	Ordered collections vs. deduplication and membership checking.
Stacks	LIFO (Last-In, First-Out) operations via push and pop.	Function call stacks, execution context, undo mechanism.
Queues	FIFO (First-In, First-Out) operations via enqueue and dequeue.	Task processing queues, event loops, BFS traversals.
Trees	Hierarchical parent-child node relationships (e.g., BSTs).	Hierarchical representations, fast search, DOM trees.

3. Advanced Data Structures

Complex structures optimized for fast lookups, priority indexing, and relational network modeling.

Data Structure	Average Time Complexity	Key Features
Hash Tables	Lookup: $O(1)$, Insert: $O(1)$	Key-value mapping using hashing algorithms to resolve collisions.
Heaps	Find Min/Max: $O(1)$, Insert: $O(\log n)$	Binary trees maintaining min/max order for priority queue tasks.
Graphs	Traversal (BFS/DFS): $O(V + E)$	Vertices connected by edges modeling complex dynamic networks.